# Endogenous Model Uncertainty and Curiosity under Cognitive Limits

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**Abstract**

What is curiosity: what leads a person to seek the information required to explore new perspectives? Neuroscience suggests that people automatically classify a situation before choosing how to act, drawing a familiar interpretation from similar past situations (e.g., a farmer attributing a neighbor's unusually high yield to good weather). As that interpretation begins to fail, people entertain less familiar alternatives. This paper treats contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain. Each mental model pairs a prior with a view of how observable features combine to reveal the state, so weighting models changes how a person interprets a situation. After an initial signal, an agent with a costless default and latent alternatives chooses these weights and how much information to acquire. The key discipline is a neurally grounded evidence constraint that bounds motivated beliefs: an alternative's weight is capped by its Bayesian posterior. Curiosity arises endogenously as a drive to resolve the resulting model uncertainty that imposes hedging costs. The model isolates informational mechanisms for two effects central to models of associative memory. First, context effects: payoff-irrelevant cues (e.g., the blue color of a fertilizer spoon) sway decisions only when another model (e.g., one in which dosage matters and blue marks the recommended dose) makes them relevant, and the curiosity this creates drives self-correction. Persistent context effects thus reflect the information environment. Second, interference effects: rather than changing which model comes to mind, rival explanations dilute one another's support, leaving none activated and search dormant. I illustrate design principles for stimulating curiosity about new programs, pedagogies, and farming techniques using examples from development economics.

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*A mind is not a vessel to be filled, but a fire to be kindled.*

— Plutarch, *On Listening to Lectures*

*The efficiency of a human problem solver depends critically on his ability to find a representation of the problem that restricts search to a small subspace of the total problem space.*

— Allen Newell and Herbert A. Simon, *Human Problem Solving* (1972)

# 1 Introduction

What is curiosity: what leads a person to seek the information required to explore new perspectives? Neuroscience suggests that people automatically classify a situation before choosing how to act (Sanders et al. 2020; Behrens et al. 2018). Memory supplies a familiar interpretation drawn from similar past situations (e.g., a farmer attributing a neighbor's unusually high yield to good weather). As that interpretation begins to fail, people entertain less familiar alternatives (Leutgeb et al. 2005a; Wills et al. 2005). Which interpretation is in use then governs what they notice and whether they look for more information, and so whether they can correct a mistaken reading. Economic theories capture this shift only partially. Models of model switching let an agent abandon her prior when an observation is sufficiently unexpected (Ortoleva 2012) or when the Bayes factor against the incumbent crosses a fixed threshold (Ba 2026). Associative-memory models of categorization let the situation retrieve a representation by its similarity to past experience, and that representation then governs what the agent attends to and how she values what she sees (Bordalo et al. 2020, 2026). Both allow one model at a time, so attention cannot move gradually toward an alternative or rest partway between two, and both abstract from information acquisition, so they say which model governs but not what the agent goes on to look for. Theories of decision under model uncertainty do allow several models at once, but the weights on them are part of preferences, as in Klibanoff et al. (2005), henceforth KMM, or are disciplined statistically, by distance from a reference model (Hansen and Sargent 2008; Maccheroni et al. 2006; Strzalecki 2011), rather than by cognitive limits. None of these frameworks has a place for the drive to seek information that arises when a person is torn between two readings of the same situation.

This paper models contextual shifts in "mindsets" as a choice of how much model uncertainty to entertain after an initial signal. The competing models are what psychologists call mental models and neuroscientists call cognitive maps. I define a mental model as a prior over the payoff-relevant state together with a likelihood that says how observable features combine to reveal that state, so two mental models differ not only in what they expect but in what they attend to and how they read it, and weighting mental models changes how a person interprets a situation. A parent who sees a school farm and thinks "children doing chores" is using one mental model; a parent who sees the same farm and thinks "a place where my child learns scientific knowledge" is using another.[1] The agent starts from a default mental model that is costless to use and faces a menu of latent alternatives that are less precise and potentially more promising. Holding an alternative is costly, both because it takes mental effort and because a single action must then hedge across models that recommend different things, and the weight she may place on it is capped by how well the evidence supports it.[2] Once some weight is placed there, she may also pay to acquire signals. The agent thus chooses how much

[1] I use "mental model" throughout, and "cognitive map" only when referring to the neuroscientific theory itself.

[2] These features are drawn from cognitive map theory, the Nobel-recognized account of how the brain represents its environment (O'Keefe and Dostrovsky 1971; Hafting et al. 2005), which suggests that the human brain is a predictive machine that actively classifies information into different domains, and such classification is malleable upon recognizing significant prediction errors. Another related line of research is the predictive coding theory (see, for example, Rao and Ballard 1999; Clark 2013). Section 2 states the regularities the model uses.

model uncertainty to entertain through a joint decision on (i) the attention she pays to each alternative and (ii) how much information to acquire, and she acquires information largely to the extent that she has already committed attention to an alternative.

The key discipline is a neurally grounded evidence constraint, so that the agent is not simply choosing the model she would like to be true, as she can in models of motivated beliefs (Brunnermeier and Parker 2005; Bénabou and Tirole 2016; Golman et al. 2017). The weight an alternative can carry is capped by the probability a Bayesian who started out indifferent would place on it after seeing the same evidence. That bound is what the theory's own account of how the brain assigns a situation to a map leaves for deliberation, and it places the agent between two familiar benchmarks: Bayesian model averaging, in which the weights are posteriors and nothing is chosen, and optimal expectations, in which they are chosen by value and nothing caps them. She may be *less* moved by evidence than Bayes requires, but never more (Section 2.3). Two further features reinforce the discipline, and each is a formal result rather than an assumption. Because the agent must choose one action whatever mixture of mental models she holds, holding two mental models forces a compromise action, and that compromise is a cost the alternative's advantage must first pay for (Proposition 1). And a vague alternative predicts everything a little and nothing well, so the evidence favors it less than a sharp one, and wishful mental models are harder rather than easier to justify (Proposition 5).

Curiosity is then generated within the model rather than assumed. It arises from a tension between accounts that could each fit what the agent has seen, and it is defined as the value to her of resolving which of them is right at the weights she holds (Definition 1), the part of the value of information that comes from discriminating between the models rather than from refining either. Where no such tension exists there is nothing to resolve: the default is not certain, only sharper than the alternatives, so under the default alone the agent is uncertain but faces only risk within one mental model, and curiosity is zero however diffuse that model is. Once an alternative is activated the tension is real, and information is worth more the more of it she entertains (Proposition 3, Corollary 1); it is greatest when the two accounts are most nearly balanced and falls away as either prevails. This formalizes the information-gap account of curiosity (Loewenstein 1994; Golman and Loewenstein 2018) rather than replacing it, since a gap presupposes a question and here the question is the tension itself. Resolving the doubt also ends the cost of holding two mental models at once: the compromise action persists as long as the doubt does, and enough information recovers exactly its cost (Proposition 4). Starting from the default, then, information that bears on the alternatives, the features the default does not encode (Assumption 1), is sought only once an alternative is activated, and the framework ties the failure to seek information about a program not to a lack of facts but to the absence of a second model on which those facts would bear.

The model reproduces regularities of neural remapping (Section 3.8), and it links several behavioral puzzles in belief updating and information acquisition: underreaction to evidence alongside abrupt revisions, the avoidance of freely available information, and the failure to notice what a mental model does not encode (Section 3.9). Activation requires two conditions at once: the alternative must be worth more than the default, and the observation must make the default indefensible against it. Neither suffices alone. Given the alternative the evidence activates, threshold effects decide how attention shifts. Each mental model recommends an action, and holding two mental models means acting on a compromise between them. When the activated alternative calls for roughly what the default calls for, the compromise costs little and the weight tracks the evidence continuously; when it calls for something different, the compromise is the worst place to be, and the weight switches between the default and the evidentiary limit as the evidence crosses a threshold (Proposition 1). Because the bound is a Bayes factor rather than a preference, an alternative that is worth more but is not supported cannot be adopted, however large its value (Proposition 2). And when the signal supports the alternative only weakly, the weight it justifies falls short of the level at which any information is worth buying, so the alternative is held but not investigated and never gains weight (Corollary 1).

The model also isolates informational mechanisms for two effects central to models of associative memory. The first is context effects. In associative-memory models, payoff-irrelevant cues change which repre-

sentation comes to mind and so change behavior (Bordalo et al. 2026). Here such cues have limited effects. Because each mental model's attention mapping is *derived* rather than posited, all mental models are judged against the same observation, and the evidence bound splits into an encoding, a surprise, and an Occam channel (Proposition 5). A feature the default does not read can then move behavior only through the encoding channel of an alternative that reads it, and a feature no mental model reads cancels from every bound and changes nothing (Corollary 2). The blue color of a spoon used to measure fertilizer (Chandrasekhar et al. 2022) is irrelevant to a farmer who thinks dosage does not matter; it becomes informative only under a mental model in which dosage matters and blue marks the recommended dose. Where a cue does tip the balance, the curiosity it creates makes the shift self-correcting: what the agent then chooses to look at, the features about which the two models disagree, carries the shift forward if the alternative is right and reverses it if it is wrong (Proposition 8, Corollary 4). A distortion of where she looks, such as a taste for confirming evidence, can slow this correction but cannot reverse it (Remark 1). Persistent context effects therefore reflect the information environment: they point to evidence that is scarce, costly, or biased, rather than to a fixed cognitive bias.

The second is interference effects. In memory models, interference changes which representation is retrieved. Here every alternative remains available, and rival explanations interfere instead by diluting one another's support: when each of several alternatives is worth more than the default, none may be activated, because the same evidence must justify each of them and justifies none enough (Proposition 6). Information concentrated on one alternative therefore succeeds where information spread across several fails, and the failure has an exact form: however large the informational budget, spreading it leaves the most weight any alternative can receive below $\max_k L_k / \sum_j L_j < 1$, while concentrating it drives that ceiling to one (Proposition 7). Search is then dormant: with every bound below threshold the agent investigates nothing, and nothing she sees moves a bound (Corollary 4), so a first plausible explanation makes the agent curious and enough of them leave her incurious (Corollary 3). Where crowding lowers the bounds without pushing them below threshold, alternatives are activated at smaller weights, and since information about an alternative is acquired only once its own weight clears a level, divided attention can buy none (Proposition 8(i)).

I apply the model to the design of information interventions in a program of school-based agricultural education (SBAE) in Liberia (Lee 2025) that aim to draw parents' attention to the program. Here the framework offers a new lens into aspirations failures: the failure to aspire is a failure to activate new mental models. For SBAE to pay off, several things must hold together: the program must be implemented well, funding must outlast the donor, the techniques must suit local soils, students must acquire management skills as well as tasks, and teachers must change how they teach. These conditions interact, so the alternative is a conjunction the parent cannot verify from what she sees. This is unlike simple categorization problems, where the agent is held to a default by the similarity between the cues before her and her past experience (Bordalo et al. 2026). Contrast therefore matters: simply engaging parents with objective facts is less effective than interventions that make the familiar model indefensible against an alternative, especially because SBAE's success depends not on parents' initial responses but on subsequent active information acquisition (e.g., visits to schools, communication with students about farming techniques, tracking students' learning progress, and assigning plots for home projects).

The framework, with its emphasis on the default mental model, has natural applications in teaching and learning. Consider three further cases: a teacher without relevant experience views experiential methods through the lens of the traditional classroom; a student who acquires arithmetic skills outside school struggles to apply them in academic settings (Banerjee et al. 2025); and a farmer who observes a surprising yield on a neighbor's plot is paralyzed by competing hypotheses about why. In each, the framework explains why interventions fail when mindsets do not shift, and isolates what makes others succeed: weighing the old model against the new, as reflection between training sessions does for teachers, and sharpening the new model through experience, as peer study groups and teaching simulations do. Together these foster active, contextual exploration that persists once the intervention ends. The student's fragmented skills need to be recognized as one domain rather than two. The third case, being high-dimensional, further shows the value

of narrowing attention to a few hypotheses, pointing to concrete ways that science education can strengthen incentives for experimentation by teaching why a technique works and under what conditions.

**Organization.** Section 2 sets up the model, with one costless default and $K$ latent alternatives, and derives the evidence constraint. Section 3 derives the results and checks them against the neuroscientific regularities. Section 4 applies them to information interventions aimed at parents, and Section 5 to pedagogy and learning among teachers, students, and farmers. Proofs are in the Appendix.

# 2 Setup

An agent faces an unknown payoff-relevant state and holds a menu of mental models of her situation: a costless default and $K$ latent alternatives. After an initial observation, she chooses jointly (i) the attention weights she places on the alternatives, which set how much model uncertainty she entertains, and (ii) how many costly signals to acquire, that is, "active" learning. She then takes a single act that must serve every model she holds. Two limits discipline the weights: holding alternatives is cognitively costly, and no alternative may receive more weight than the evidence supports. The agent maximizes expected utility; no ambiguity aversion is assumed, and adding it only strengthens the results. The menu is taken as fixed throughout. The structure rests on three insights from neuroscience, specifically cognitive map theory, which originally dealt with spatial navigation but has recently been extended to learning about conceptual ideas, memory mapping, and social relationships (Constantinescu et al. 2016; Behrens et al. 2018; Whittington et al. 2020):

- **Default, familiar preconception (auto-pilot mode).** Agents begin with a costless default prior that is more precise than the alternatives (Kool et al. 2010; Shenhav et al. 2013).
- **Costly activation of latent mental models.** Activating an alternative prior requires cognitive effort, modeled as a convex cost, and holding several alternatives at once is more costly than holding each alone (Shenhav et al. 2017; Lieder and Griffiths 2020).
- **Complementarity between activation and information acquisition.** The more an agent commits to a new prior, the more valuable signal acquisition becomes; conversely, signals raise the value of the new prior.

## 2.1 Mental models, signals, and timing

Let $\theta \in \mathbb{R}$ denote an unknown payoff-relevant state: the return to a course of action whose value the agent does not know. The agent has von Neumann–Morgenstern preferences represented by $u(a,\theta)$ over acts $a \in \mathscr{A} \subseteq \mathbb{R}$ and states $\theta$, and $\mathbb{E}_\mu\, u(a,\theta)$ denotes expected utility under a belief $\mu$.

**Mental models.** The agent holds a finite menu of mental models of the situation she faces, indexed by $k \in \{0,1,\ldots,K\}$. Index $k=0$ is the *default* model, which she has applied many times before and which she operates on costlessly; indices $k=1,\ldots,K$ are *latent alternatives*, which she may or may not bring to bear. The menu is fixed: subsequent signals reweight and refine the models on it rather than introducing new ones. Taking it as given is where an associative account of memory enters: what an agent can entertain is what the situation has brought to mind, so retrieval determines the menu and the problem studied here begins once retrieval has happened. Nothing below requires the menu to be the same across situations or agents, only that the weight placed on whatever is available be disciplined by the evidence. A mental model has two aspects. The first is a *prior* over the state: model $k$ asserts $\theta \sim \mathcal{N}(\mu_k, \sigma_k^2)$ with precision $\tau_k \equiv 1/\sigma_k^2$, and the default is the most precise on the menu, $\tau_0 > \tau_k$ for every $k \geq 1$, because it has been applied most often. The means are left unordered. The second aspect is a *generative process* that says how the observable features combine

to reveal the state, which determines both what the model attends to and how it interprets what it sees. The agent observes a raw feature vector $x \in \mathbb{R}^d$, such as the content and ordering of a message or what is seen on a visit, with the scalar case recovered at $d = 1$. As the cognitive-map theory treats it, each model is a complete generative process for those features:

$$\theta \sim \mathcal{N}(\mu_k, \sigma_k^2), \qquad x \mid \theta \sim \mathcal{N}(\theta\, b_k,\, \Sigma), \qquad \tau_k := 1/\sigma_k^2, \qquad k = 0, 1, \ldots, K, \tag{1}$$

where $b_k \in \mathbb{R}^d \setminus \{0\}$ is the mental model's loading vector and $\Sigma \succ 0$ is a common noise covariance.[3] Each alternative is structurally distinct from the default and from the other alternatives, in that it carries its own loading $b_k$.

The attention mapping is not an extra primitive. It is what each mental model's own inference problem implies.

**Lemma 1** (Attention mappings are derived)**.** *Under* (1)*, the posterior over $\theta$ given $x$ depends on $x$ only through the scalar*

$$s_k \;=\; \alpha_k' x, \qquad \alpha_k \;=\; \frac{\Sigma^{-1} b_k}{b_k' \Sigma^{-1} b_k}, \tag{2}$$

*and $s_k \mid \theta \sim \mathcal{N}(\theta, \nu_k^2)$ with $\nu_k^2 = (b_k' \Sigma^{-1} b_k)^{-1}$. The posterior is Gaussian with precision $P_k(0) = \tau_k + \nu_k^{-2}$ and mean $m_k = \big(\tau_k \mu_k + \nu_k^{-2} s_k\big)/P_k(0)$.*

Each mental model thus compresses the observable features into a single number $s_k$ carrying everything it needs to know about $\theta$, and the weights in that compression, $\alpha_k$, are its attention mapping, the combination of features its own model treats as informative. When $b_j \neq b_k$ two models read the same raw observation differently, and features diagnostic under one may be ignored by the other, so shifting weight between them changes how the agent interprets the same situation; when $b_j \propto b_k$ they attend identically and differ only in $(\mu_k, \tau_k)$. Proposition 5 shows that the distinction has observable content, which gives a formal edge to what is otherwise verbal: two priors differing only in their means are *not* two mental models in any sense that matters here.

**Signals.** After observing $x$, the agent may acquire $n \geq 0$ subsequent signals at marginal cost $c$ per signal, each of the form $y_i = \theta + \eta_i$, $\eta_i \sim \mathcal{N}(0, \nu_y^2)$.

**Attention weights.** The agent chooses a vector of attention weights $\mathbf{w} = (w_1, \ldots, w_K)$ on the alternatives, with $w_k \geq 0$ and $\sum_{k \geq 1} w_k \leq 1$, the remainder $w_0 = 1 - \sum_{k \geq 1} w_k$ resting on the default; and she chooses the number of signals $n$. Timing is:

(1) $x$ is realized and observed.

(2) The agent chooses the weights $w_k \in [0, \overline{W}_k(x)]$ on the alternatives and the number of costly signals $n \in \{0, 1, 2, \ldots\}$.

(3) The $n$ signals realize.

(4) The agent chooses a single act $a$, evaluated at the weights chosen in step (2).

Section 3.4 lets the acquired signals update the weights before step (4); the results below concern the weights chosen at step (2). Step (4) fixes the structure of the problem. The agent chooses a single act under model uncertainty, so the maximization runs over one $a$ evaluated at the aggregated weights, as in (4). Aggregating model-contingent maxima $\max_a \mathbb{E}_k u(a, \theta)$ would instead evaluate the position of an agent who

[3] Allowing $\Sigma_k$ to differ across mental models changes nothing conceptually and adds a fourth term to Proposition 5. A common $\Sigma$ isolates the channels of interest.

*knows* which mental model applies. Placing the maximum outside the sum over models is what gives the disagreement between mental models its bite, and it generates the hedging cost of Section 2.4.

## 2.2 Preferences, costs, and the derived objective

The agent evaluates an act by its expected utility under each mental model and aggregates across models with the weights. To nest attitudes toward model uncertainty, the aggregation is written through a strictly increasing $\varphi : \mathbb{R} \to \mathbb{R}$. The baseline sets $\varphi(U) = U$: the agent maximizes expected utility under the mixture of models and has no ambiguity aversion as such. A strictly concave $\varphi$ adds ambiguity aversion. It raises activation thresholds and directs curiosity toward evidence that discriminates between models, so it strengthens the results rather than drives them.

Holding alternatives active imposes a cognitive cost,

$$C(\mathbf{w}) \;=\; \frac{\gamma}{2}\sum_{k\geq 1} w_k^2 \;+\; \eta \sum_{1\leq j<k\leq K} w_j w_k, \qquad \eta > \gamma > 0. \tag{3}$$

The cost has three properties. First, operating under the default alone is costless. Second, it is convex: each additional unit of attention to an alternative requires progressively greater inhibition of the dominant default and progressively greater suppression of interference from other active alternatives.[4] Third, because $\eta > \gamma$ the cost is superadditive across alternatives: holding two alternatives at once costs strictly more than the sum of holding each alone. When a single alternative $k$ is held, (3) is simply $(\gamma/2)w_k^2$. A count-based cost such as $\gamma(m-1)^2$ in the number $m$ of active models, by contrast, makes the problem a mixed integer program with no first-order conditions in the weights. The agent weighs the value function against this cost and the information-acquisition cost $cn$, selecting $\mathbf{w}$ and $n$ to maximize the objective below subject to the evidence constraints $0 \leq w_k \leq \overline{W}_k(x)$ of Section 2.3 and $n \in \mathbb{Z}_{\geq 0}$.

The cost (3) is not an information cost. A relative-entropy penalty for departing from a reference distribution over models, as in robustness and rational inattention (Hansen and Sargent 2008; Sims 2003), *rewards* spreading weight across alternatives, since entropy favors dispersion. The interference term $\eta > \gamma$ does the opposite, and it is what the working-memory capacity limit contributes (Cowan 2001; Bays and Husain 2008); it drives the concentration result of Section 3.1 and the fragmentation result of Section 3.6.

Writing $\mathbb{E}_{k,n}$ for the expectation under mental model $k$'s posterior after $x$ and $n$ signals, and $\mathbb{E}^y$ for the expectation over signal realizations taken at step (2), the derived objective is

$$V(\mathbf{w}, n; x) \;=\; \mathbb{E}^y\Big[\max_{a\in\mathscr{A}} \sum_{k=0}^{K} w_k\, \varphi\big(\mathbb{E}_{k,n} u(a,\theta)\big)\Big] \;-\; C(\mathbf{w}) \;-\; cn. \tag{4}$$

Equation (4) is derived from vNM utility, the aggregation across models, and convex costs. The agent is not maximizing a cardinal value function for its own sake; she is choosing how much model uncertainty to entertain, jointly through the weights $\mathbf{w}$ and the signal count $n$, subject to the cognitive cost and the evidence constraints. Nothing in (4) is primitive except $u$, $\varphi$, $\gamma$, $\eta$, and $c$.[5]

[4] The convex form is the natural reduced-form summary of two well-documented neural constraints. Working-memory capacity is severely limited, to roughly three to four items (Cowan 2001), and the precision with which items are held drops nonlinearly rather than linearly as memory load increases (Bays and Husain 2008). And tracking competing explanations is more demanding than tracking items, because each hypothesis carries its own predictions that must be updated against incoming evidence.

[5] What the analysis uses is an explicit second-order belief over mental models that updates by Bayes' rule (Section 3.4). Propositions 1 through 8 set $\varphi$ linear, so they hold for an expected-utility agent; the curvature of $\varphi$ is used only in Proposition A.2. Representations without a second-order belief, such as maxmin (Gilboa and Schmeidler 1989), variational or confidence preferences, have no weight to cap and no counterpart to activation.

## 2.3 The evidence constraint

The evidence constraint means that the weight on an alternative cannot exceed the evidentiary support for that alternative: it is a constraint on likelihood, not on value, and it binds what the agent may believe rather than what she may observe. It is cognitive in origin even though its content is evidentiary, because it is what the brain's own assignment of a situation to a model leaves for deliberation; the cognitive cost of Section 2.4 is the second limit, and the two together are the cognitive limits of the title. This has a neural basis: categorical activation is justified by evidence, not by preference; an agent cannot adopt a prior the data do not support, no matter how attractive its mean is (Sanders et al. 2020). Let $L_k(x)$ be the probability that mental model $k$ assigned, before the fact, to the features that were actually observed, its marginal likelihood, and let $\Lambda_k(x) := L_k(x)/L_0(x)$ be the Bayes factor of alternative $k$ against the default: how much better alternative $k$ accounts for what was seen than the default does. Define

$$\overline{W}_k(x) \;:=\; \frac{L_k(x)}{L_0(x)+\sum_{j=1}^{K} L_j(x)} \;=\; \frac{\Lambda_k(x)}{1+\sum_{j=1}^{K} \Lambda_j(x)}, \qquad k=1,\dots,K. \tag{5}$$

When $x$ is consistent with the default, $L_0$ dominates and every $\overline{W}_k \approx 0$, foreclosing meaningful weight on any alternative. When $x$ falls in the tails of the default, the ratios reverse and the bounds of the alternatives that account for $x$ rise. The right-hand side of (5) is exactly the posterior probability a Bayesian with a uniform prior over the $K+1$ mental models would assign to alternative $k$ after seeing $x$, so the constraint $w_k \le \overline{W}_k(x)$ reads: *the agent may not be more convinced of an alternative than the evidence justifies*. The constraint is one-sided. She may remain at $\mathbf{w}=0$ however strong the evidence, which is the sense in which the default is sticky; she may never exceed $\overline{W}_k$, which is the sense in which the model is not one of motivated beliefs. A default-favoring prior over mental models rescales $L_0$ in (5) and changes nothing below. Two features of the bound matter for what follows. The denominator includes every alternative, not just the focal one, so each alternative's admissible weight depends on the evidentiary support enjoyed by every other alternative the agent might entertain; Section 3.6 draws out the consequences. And when only one alternative is on the menu the bound is $\Lambda_k/(1+\Lambda_k)$, the posterior probability in a two-model comparison.

**Where the constraint comes from.** The bound is not imposed from outside; it is what the brain's automatic assignment of a situation to a map leaves for deliberation. Sanders et al. (2020) show that hippocampal remapping behaves like Bayesian inference over which context the animal is in: the map that becomes active is the one favored by the posterior over contexts given the evidence. Taking that as the automatic process, after $x$ it assigns to model $k$

$$p_k(x) \;=\; \frac{q_k L_k(x)}{\sum_{j=0}^{K} q_j L_j(x)}, \tag{6}$$

where $q$ is the prior the automatic process carries. Deliberation can hold a model only to the extent the automatic process has made it available, $w_k \le p_k(x)$; with a uniform $q$ this is (5), and a default-favoring $q$ rescales $L_0$. The one-sidedness has its own counterpart: a well-learned map is an attractor, so remaining in it requires no evidence and it snaps back after a brief disturbance (Wills et al. 2005; Jezek et al. 2011), whereas leaving it requires evidence. This is a reading of the neuroscience rather than a derivation from neural dynamics; what it commits the model to is the property the results use, that deliberate activation is bounded by posterior probability.

**Two limiting cases.** The constraint places the agent between two benchmarks the literature treats separately (Figure 1). Were holding models costless, she would set every weight at its cap: that is Bayesian model averaging, in which weights are posterior probabilities and nothing is chosen. Were there no cap, she would set the weights by the value of the models they favor, tempered only by the costs: that is the structure of

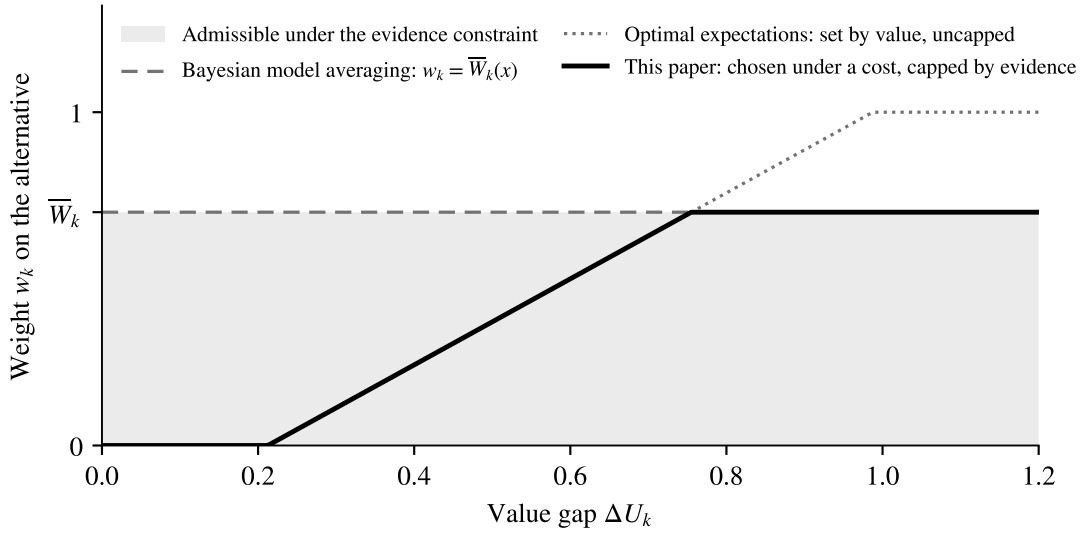



Figure 1: The agent between two benchmarks.

*Notes.* The line is the weight $w_k$ on an alternative. Bayesian model averaging sets it at $\overline{W}_k(x)$; optimal expectations set it by value, uncapped; the agent here chooses it under a cost, within the shaded region the evidence justifies.

optimal expectations (Brunnermeier and Parker 2005), which Proposition 2 states exactly. The agent lives on the segment between them: her weights are chosen, which the averager's are not, and bounded by evidence, which the optimist's are not. Everything the paper adds (activation as a choice, curiosity as its consequence, fragmentation, and the design principles) is about that segment.[6]

Every signal the agent encounters triggers two distinct processes in parallel. Under *within-prior Bayesian updating*, the posterior over $\theta$ for each active prior $k$ updates by Bayes' rule; this is automatic and requires no allocation choice, and its effect is to refine the precision of the posterior within whichever mental models the agent has activated. Under *across-prior weight reassignment*, the same signal provides evidence about which prior better generated the observation, the agent uses it to reassess $\mathbf{w}$, and its effect is to determine the activation of each mental model. The evidence bounds (5) are the discipline on the second process.

## 2.4 The cost of holding several mental models

A single act must serve whatever mixture of mental models the agent holds. The *hedging cost* of holding weights $\mathbf{w}$ is the value lost because the agent must choose one act rather than the act each mental model recommends:

$$A(\mathbf{w}) := \underbrace{\sum_{k=0}^{K} w_k \varphi\big(U_k\big)}_{\text{value of knowing the mental model}} - \underbrace{\max_{a} \sum_{k=0}^{K} w_k \varphi\big(\mathbb{E}_k u(a,\theta)\big)}_{\text{value of one act under model uncertainty}}, \tag{7}$$

[6] Given the weights, the agent's beliefs about $\theta$ are the mixture $\sum_k w_k \pi_k$, a single distribution, and with $\varphi$ linear the value of information is the expected-utility value under that distribution. The two-model language is therefore not needed to describe what she believes; it is needed to describe what governs it. A Bayes factor exists only between identified components, so the cap cannot be stated for a single free distribution; the cognitive and hedging costs are functions of the weights, and the hedging cost is the expected value of perfect information about the component index, which a single prior has no counterpart to; and the updating of Section 3.4 reweights the components by their likelihood ratio, which is not the same as updating the mixture as one prior. The paper's content lives in these three places, and none of them survives the collapse to a single prior.

where $U_k := \max_a \mathbb{E}_k u(a,\theta)$ is the model-contingent value. The hedging cost is nonnegative, zero when all weight rests on one mental model or when every model receiving weight induces the same act, and concave in $\mathbf{w}$ (Lemma A.1 in the Appendix). It is the only force in the model that makes holding several mental models intrinsically uncomfortable, independently of any cognitive cost, and it is that force which keeps endogenous weights from collapsing into wishful thinking. The next subsection computes $A$ in closed form.

### 2.5 A closed-form specification

Let the act be a scale of engagement $a \in \mathbb{R}$ with

$$u(a,\theta) \;=\; a(\theta-\kappa) - \tfrac{\lambda}{2}a^2, \qquad \lambda > 0, \tag{8}$$

where $\kappa$ is the marginal cost of engagement and $\lambda$ an adjustment cost. Costly signals are conditionally i.i.d. draws $y_i = \theta + \eta_i$, $\eta_i \sim \mathcal{N}(0, \nu_y^2)$, so mental model $k$'s posterior precision after $x$ and $n$ signals is $P_k(n) = \tau_k + \nu_k^{-2} + n\nu_y^{-2}$ and, because today's belief is the average of the beliefs one might hold tomorrow, the variance of the step-(4) posterior mean, viewed from step (2), is

$$\psi_k(n) \;=\; \frac{1}{P_k(0)} - \frac{1}{P_k(n)}, \qquad \psi_k' > 0, \quad \psi_k'' < 0. \tag{9}$$

Under (8), the model-contingent value is available in closed form:

$$U_k(x,n) \;=\; \frac{(m_k-\kappa)^2 + \psi_k(n)}{2\lambda}, \qquad \frac{\partial U_k}{\partial n} \;=\; \frac{1}{2\lambda\,\nu_y^2 P_k(n)^2} \;>\; 0, \tag{10}$$

strictly increasing and strictly concave in $n$. I assume that the default remains the most precise model after the observation, $P_0(0) > P_k(0)$ for every $k \geq 1$, which holds whenever $\tau_0 > \tau_k$ and the loadings are equally strong, $\nu_k = \nu_0$. Then $P_k(n) < P_0(n)$ for every alternative, and signals are worth strictly more under any alternative than under the default at every $n$.

In the baseline $\varphi(U) = U$; Section 3.2 notes what concavity changes. Then (7) evaluates to

$$A(\mathbf{w}) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(m)}{2\lambda} \;=\; \frac{1}{2\lambda}\sum_{k=0}^{K} w_k\,(m_k - \bar{m})^2, \qquad \bar{m} := \sum_{k=0}^{K} w_k m_k, \tag{11}$$

the weighted variance of the model-implied posterior means, divided by the adjustment cost. It is zero when the agent is certain of one mental model and larger the more the mental models she holds disagree about $\theta$. When a single alternative $k$ is held, (11) reduces to

$$A \;=\; w_k(1-w_k)\,D_k(x), \qquad D_k(x) \;:=\; \frac{(m_k - m_0)^2}{2\lambda}, \tag{12}$$

which is maximal at $w_k = 1/2$. Call $D_k(x)$ the *disagreement* between alternative $k$ and the default. It is not a parameter of the agent: $m_k$ and $m_0$ are posterior means, each formed by reading the observation through its own attention mapping, so $D_k(x)$ measures how far apart the two mental models' recommended actions are *given the signal*, and the same agent faces a different $D_k$ for different observations and for different alternatives. Its $w(1-w)$ shape is the same one that the source function of Abdellaoui et al. (2011) and Baillon et al. (2018) attaches to perceived ambiguity, which is what makes interior weights, rather than a diffuse single prior, the object those indices are picking up. The objective becomes

$$V(\mathbf{w},n) \;=\; \sum_{k=0}^{K} w_k U_k(x,n) \;-\; A(\mathbf{w}) \;-\; C(\mathbf{w}) \;-\; cn, \tag{13}$$

and, when a single alternative $k$ is held at weight $w_k$,

$$V(w_k,n) \;=\; (1-w_k)\,U_0(x,n)+w_k\,U_k(x,n) \;-\; w_k(1-w_k)D_k(x) \;-\; \tfrac{\gamma}{2}w_k^2 \;-\; cn. \tag{14}$$

# 3 Results

This section states the properties of mindset shifts that the model delivers: shifts happen one alternative at a time; an alternative is activated only when the evidence makes the default indefensible against it, and threshold effects then govern how attention shifts; information is sought only once an alternative is activated, and the doubt it creates is resolved in one of two ways; what moves the evidence bounds; and why evidence divided across several alternatives activates none. The design principles these properties imply are collected in Section 4.

## 3.1 One alternative at a time

Before characterizing activation it is worth asking how the agent spreads attention across alternatives, because the answer simplifies everything that follows. For a given total attention off the default, the cost (3) is uniquely minimized by placing all of it on a single alternative, and any split pays an interference premium of $(\eta-\gamma)\sum_{j<k}w_jw_k$ (Lemma A.2 in the Appendix). The agent's problem therefore has a built-in preference for entertaining one alternative at a time, the endogenous counterpart of a restriction imposed by assumption elsewhere in the literature (Schwartzstein and Sunderam 2021; Gabaix 2014). On the face of the simplex where a single alternative $k$ is held, the objective is (14): the problem is a comparison between alternative $k$ and the default, with the disagreement $D_k(x)$, the value gap $\Delta U_k := U_k(x,n)-U_0(x,n)$, and the evidence bound $\overline{W}_k(x)$ all computed from the observation. The agent's choice of *which* alternative to entertain is then the choice of the face on which (14) is highest, and the observation makes that choice for her through the same three objects: it decides which alternatives clear their evidence bounds, how much each is worth, and how far each stands from the default. Mindset shifts are therefore one-alternative-at-a-time affairs, and the next subsections characterize them on a given face; Section 3.6 returns to the interaction between alternatives.

## 3.2 Activation: the value and evidence conditions

Because the default is precise, its posterior mean is anchored to $\mu_0$ and responds modestly to the data: a favorable observation tends to be absorbed into noise rather than shifting the value of the default categorization.

Activation of alternative $k$ requires two conditions at once. The *value condition* is that the value gap $\Delta U_k$ clear a threshold $T_k$, so that the alternative is worth enough to pay for holding both models. The *likelihood condition* is that $x$ be inconsistent enough with the default to make $\overline{W}_k(x)$ nontrivial. Neither suffices: the value gap net of the threshold gives the desired weight, and $\overline{W}_k(x)$ caps how much of it is admissible. Which form the value condition takes depends on the curvature of (14) in $w_k$, namely $2D_k-\gamma$: the hedging cost is concave and destabilizes interior weights, while the cognitive cost is convex and stabilizes them, so $T_k$ is stated regime by regime.

**Proposition 1** (Activation)**.** *Fix $n$ and an alternative $k$ held on its own, and write $D_k = D_k(x)$ and $\overline{W}_k = \overline{W}_k(x)$. Activation of $k$ requires a value condition and an evidence condition jointly,* $\min\{\Delta U_k - T_k,\ \overline{W}_k\} > 0$*, where the threshold $T_k$ and the form of the optimum depend on the curvature $2D_k-\gamma$.*

*(a)* (Graded regime, $\gamma > 2D_k$.) *$T_k = D_k$, $V$ is strictly concave in $w_k$, and*

$$w_k^* \;=\; \mathrm{med}\Big\{0,\ \frac{\Delta U_k - D_k}{\gamma - 2D_k},\ \overline{W}_k\Big\}.$$

*Activation occurs if and only if $\Delta U_k > D_k$, and $w_k^*$ is continuous and increasing in $\Delta U_k$.*

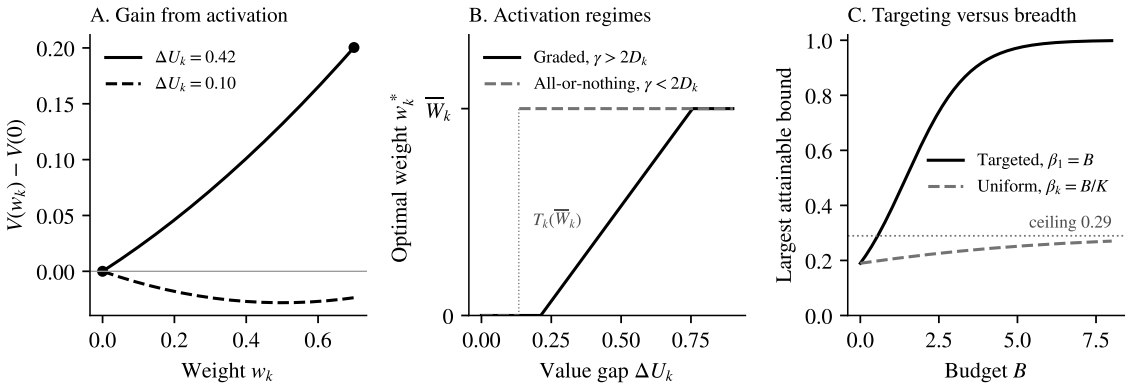



Figure 2: Activation regimes and the ceiling on breadth.

*Notes.* Panels A and B set $D_k = 0.2125$, $\overline{W}_k = 0.70$, with $\gamma = 1.20 > 2D_k$ (graded) and $\gamma = 0.20 < 2D_k$ (all-or-nothing). Panel A plots $V(w_k) - V(0)$ from (14) at $\Delta U_k = 0.42$ (solid) and 0.10 (dashed), with markers at $w_k^*$; in the convex regime a small value gap makes partial activation worse than the default. Panel B plots the optimal weight against the value gap: continuous above $\Delta U_k = D_k$, a step at $T_k(\overline{W}_k) = 0.134$. Panel C plots the largest attainable bound against a budget $B$ for $K = 4$ with $(L_0; L_1, \ldots, L_4) = (1.0; 0.55, 0.50, 0.45, 0.40)$ under targeted ($\beta_1 = B$) and uniform ($\beta_k = B/K$) allocations; the dashed line is the ceiling 0.29 of Proposition 7.

*(b)* (All-or-nothing regime, $\gamma < 2D_k$.) *$V$ is strictly convex in $w_k$, so $w_k^* \in \{0, \overline{W}_k\}$, and $w_k^* = \overline{W}_k$ if and only if*

$$\Delta U_k \;>\; T_k \;=\; T_k(\overline{W}_k) \;:=\; D_k(1 - \overline{W}_k) + \tfrac{\gamma}{2}\overline{W}_k \;<\; D_k.$$

*The threshold is strictly below $D_k$: when the alternative and the default disagree enough, a decisive shift can pay even though no partial shift would. The threshold $T_k$ is strictly decreasing in $\overline{W}_k$, and at the threshold $w_k^*$ jumps discontinuously from 0 to $\overline{W}_k$.*

*(c)* (Knife edge, $\gamma = 2D_k$.) *$T_k = D_k$, $V$ is linear in $w_k$, and $w_k^* = \overline{W}_k$ if $\Delta U_k > D_k$, $w_k^* = 0$ otherwise.*

*Both $\overline{W}_k(x)$ and $D_k(x)$ are functions of the observation, so the observation determines which alternatives can be activated and, for each, whether the weight on it moves continuously or in switches.*

Figure 2 illustrates. Proposition 1 is the formal statement of the first result. The evidence condition is the first part: an alternative whose bound (5) the observation leaves near zero cannot be activated whatever its value, and because each bound depends on how the observation is read and not only on what it contains (Section 3.5), contrast decides which alternative is activated. The value condition is the second, and its threshold depends on a regime the observation also decides. A signal that alternative $k$ reads as calling for roughly what the default calls for leaves $m_k$ near $m_0$, so $D_k$ is small and the agent is in the graded regime; a signal it reads as calling for something different pushes $m_k$ away from $m_0$ and she is in the all-or-nothing regime. The same agent can therefore adjust continuously after one observation and switch after the next. Note also that activation requires more than a value advantage: in the graded regime the condition is $\Delta U_k > D_k$, not $\Delta U_k > 0$, so alternatives that disagree sharply with the default are *harder* to activate partially.

The two regimes are the counterparts of the gradual and abrupt transitions neuroscientists observe when the neurons encoding an animal's situation meet discrepant evidence: *rate remapping*, in which cells keep their firing fields and change only their rates (Leutgeb et al. 2005a), and *global remapping*, the all-at-once switch to an independent code (Wills et al. 2005; Leutgeb et al. 2005b; Colgin et al. 2008) that Sanders et al. (2020) interpret as Bayesian inference about which situation the animal is in. The model generates the dichotomy rather than assuming it, though nothing here establishes that its classifying variable is what those experiments manipulate.

The dichotomy does not depend on the quadratic payoff, the Gaussian structure, or a linear $\varphi$: the objective

is a difference of convex functions, and which term has the greater curvature decides whether the weight moves continuously or sits at a corner (Remark A.2 in the Appendix). The constraint is what separates the model from one of motivated beliefs, and the separation can be stated exactly.

**Proposition 2** (What the evidence constraint buys)**.** *Remove the constraint, so that $w_k \in [0,1]$ on the face where alternative k is held. Then the optimal weight is $w_k^* = \text{med}\{0, (\Delta U_k - D_k)/(\gamma - 2D_k), 1\}$ in the graded regime and $w_k^* \in \{0,1\}$ with threshold $\Delta U_k > \gamma/2$ in the all-or-nothing regime. In either regime $w_k^*$ is non-decreasing in the value gap $\Delta U_k$ and independent of the likelihood ratio $L_k(x)/L_0(x)$: an alternative that the observation contradicts arbitrarily strongly is adopted in full whenever it is worth enough. With the constraint, $w_k^* \leq \overline{W}_k(x)$, and $\partial w_k^*/\partial \Delta U_k = 0$ once the bound binds: value cannot buy weight the evidence does not justify.*

Without the cap the agent chooses her categorization for what it delivers, as beliefs are chosen in optimal expectations (Brunnermeier and Parker 2005; Bénabou and Tirole 2016); the hedging and cognitive costs temper how far she goes but not what she goes toward.[7] The cap makes evidence, not value, the gatekeeper (Figure 1), and it is the only feature of the model that does so.

A consequence is that isolated flukes do not produce small durable weights: in the all-or-nothing regime an observation that falls short of the threshold leaves the weight at zero exactly, and in the graded regime it produces a weight too small to prompt the information that would let it grow (Corollary A.1 in the Appendix). With a strictly concave $\varphi$, ambiguity aversion adds a second $w_k(1-w_k)$ penalty, so Proposition 1 goes through with $D_k$ replaced by a larger $\tilde{D}_k$ and the all-or-nothing regime becomes *more* likely the more ambiguity averse the agent (Remark A.1 in the Appendix).

### 3.3 Complementarity: curiosity as the drive to resolve endogenous model uncertainty

This subsection turns to the second half of the joint decision, information acquisition. Curiosity is not left as a description. The hedging cost already defined is, by Proposition 4(b) below, exactly the value of learning which mental model is right at the weights the agent holds, so it can serve as the definition.

**Definition 1** (Curiosity)**.** At weights $\mathbf{w}$, *curiosity* is the value of resolving which mental model is right,

$$A(\mathbf{w}) \;=\; \frac{\text{Var}_{\mathbf{w}}(m)}{2\lambda},$$

the component of the value of information that comes from discriminating between the models rather than from refining either.

Three properties follow from results stated elsewhere rather than from assumption. Curiosity is zero at $\mathbf{w} = \mathbf{0}$, so an agent under the default alone is not curious however diffuse that model is; on a face it equals $w_k(1-w_k)D_k$ and so rises with the weight placed on the alternative over the relevant range; and Proposition 4 separates it from the refinement of either model, which is why signals that discriminate count and signals that improve both do not. What the agent then looks at is settled by Proposition 8. Curiosity thus coincides with the hedging cost of Section 2.4: the cost of doubt and the value of resolving it are one object, which is why activation and acquisition move together: information is worth more the more of that doubt the agent entertains, and none is sought under the default alone.

Definition 1 is meant as a formalization of the information-gap account rather than a departure from it. In that account curiosity is the aversive state produced by awareness of a gap between what one knows and what one wants to know (Loewenstein 1994), and it is modeled through preferences over beliefs, with attention to

[7] The correspondence is structural rather than motivational: there the agent chooses any distribution and gains anticipatory utility from optimism, here she chooses among categories she already holds and gains nothing from holding them. The menu restricts her to motivated categorization rather than motivated belief; the cap restricts her further, to categories the evidence supports.

a question taken as given (Golman and Loewenstein 2018). Three correspondences are close. A gap requires a question, and here the question does not exist until an alternative is activated, so the definition supplies the step that account assumes. Intensity is inverted-U in knowledge there, and $w_k(1-w_k)D_k$ is inverted-U in the weight here, greatest when the agent is least sure which model applies and vanishing at either extreme. And curiosity is directed at a particular question there, while here Proposition 8 says which observations it sends the agent to look for. The restriction is that $A$ values only information that could change what the agent does, so the appetite for an answer that bears on nothing lies outside it; that restriction follows from the design problem, since an intervention can put an alternative in front of someone but cannot install an appetite for information in general. Nor is $A$ the value of information as such: the refinement component in Proposition 4 is available under a single model and is not curiosity in the sense used here.[8] The formal statement is a complementarity between $w_k$ and $n$.

**Proposition 3** (Complementarity)**.** *The effect of an additional signal on the return to attention to alternative k, the cross-partial of* (4)*, is*

$$\frac{\partial^2 V}{\partial w_k \, \partial n} \;=\; \varphi'(U_k)\frac{\partial U_k}{\partial n} - \varphi'(U_0)\frac{\partial U_0}{\partial n},$$

*which is strictly positive if and only if* $\big(P_k(n)/P_0(n)\big)^2 < \varphi'(U_k)/\varphi'(U_0)$*. Each of the following is sufficient:* (i) $\varphi$ *linear;* (ii) $U_k \le U_0$*;* (iii) *Assumption 1 below. When the inequality holds,* $(w_k, n)$ *are complements:* $w_k^*$ *is nondecreasing in n and* $n^*$ *is nondecreasing in* $w_k$*, and the first-order condition in n is* $\sum_{j=0}^{K} w_j \, \varphi'(U_j) \, \partial_n U_j = c$*.*

Complementarity is a condition rather than an identity: signals are worth more under the diffuse model, which is what $P_k < P_0$ says, and this must outweigh the curvature of $\varphi$. The sharper claim that $n^* = 0$ whenever $\mathbf{w}^* = 0$ needs one further assumption.

**Assumption 1** (Alternative-specific signals)**.** The costly signals load on features the default mental model does not encode: $\partial U_0/\partial n = 0$.

Program-specific information, such as pass rates from the school farm or an explanation of why a technique works, is information for which the default has no slot. Under Assumption 1 complementarity is unconditional and the marginal benefit of a signal at $\mathbf{w} = 0$ is exactly zero, so $n^* = 0$: there is no point acquiring detail about a model that has not been opened. Curiosity is thus absent under a single mental model, however diffuse its prior, and grows with the attention the alternative receives. Without Assumption 1 the comparative static $\partial n^*/\partial w_k \ge 0$ survives but the corner result does not. The information side is then also available in closed form.

**Corollary 1** (Closed-form information acquisition)**.** *Under Assumption 1 with* $\varphi$ *linear, and treating n as continuous, the optimal number of signals when alternative k is held on its own at weight* $w_k$ *is*

$$n^*(w_k) \;=\; \nu_y^2 \Big[ \Big( \frac{w_k}{2\lambda \nu_y^2 c} \Big)^{1/2} - P_k(0) \Big]^+,$$

*which is zero for* $w_k \le 2\lambda \nu_y^2 c \, P_k(0)^2$ *and thereafter increasing and concave in* $w_k$ *and decreasing in c. Together with Proposition 1, the optimum* $(w_k^*, n^*)$ *is the fixed point of two explicit best responses.*

Corollary 1 has an extensive margin: no signal is bought until the weight on the alternative exceeds $2\lambda \nu_y^2 c \, P_k(0)^2$, a threshold that rises with the cost of signals and with the alternative's own precision. Together

[8] Given the weights, the value of information here is an expected-utility object: the value of refining the mixture belief plus the expected value of perfect information about which model is right (Proposition 4). What is not an expected-utility object is activation, the choice of the weights, and it is activation that makes curiosity switch on, scale with the weight, and be absent under the default. A Bayesian who holds two models is always curious; the agent here is curious only once she has chosen to hold a second one.

with Proposition 1 this yields three paths after an observation. When the observation supports the alternative it activates only weakly, the weight it justifies falls short of the threshold: the alternative is held but not investigated, stays imprecise, and never gains weight. When the observation supports the alternative more strongly and the alternative recommends much the same action as the default, the weight tracks the evidence continuously and information is bought as it rises. When the alternative recommends a different action, the weight switches to the evidentiary limit and information follows. Which path the agent takes is decided by the observation, through which alternative it activates and how strongly.

### 3.4 Updating: how the doubt is resolved

The timing laid out in Section 2.1 evaluates the act at the weights chosen in step (2). This subsection lets the acquired signals update those weights, which closes the loop described in the previous subsection: attention prompts information, and information moves attention. Write $\Lambda_k^y(y^n)$ for the likelihood ratio of the $n$ acquired signals under alternative $k$ against the default, each mental model evaluated at its own step-(2) posterior, so that the predictive probability model $k$ assigned to the signals that were realized is what enters. After the signals the weights and the evidence bounds update by Bayes' rule,

$$w_{k,n} \;=\; \frac{w_k\,\Lambda_k^y(y^n)}{w_0+\sum_{j=1}^K w_j\,\Lambda_j^y(y^n)}, \qquad \overline{W}_{k,n} \;=\; \frac{\Lambda_k(x)\,\Lambda_k^y(y^n)}{1+\sum_{j=1}^K \Lambda_j(x)\,\Lambda_j^y(y^n)}, \tag{15}$$

and the act in step (4) is evaluated at $\mathbf{w}_n$. This is the recursive update of the second-order prior in Klibanoff et al. (2009), applied to weights the agent has chosen rather than inherited; the within-model posteriors update as before. When a single alternative $k$ is held, (15) reduces to $w_{k,n} = w_k\Lambda_k^y/(1-w_k+w_k\Lambda_k^y)$. Updating preserves the discipline: the odds of every alternative against the default never exceed its Bayes factor, before or after the signals, and the weights are a martingale under the mixture, so acquiring signals cannot be expected to move attention in either direction, only to resolve it (Lemma A.3 in the Appendix).

The value of the signals can now be computed with the weights moving. Let $\bar{m}_n := \sum_{k=0}^K w_{k,n}\,m_{k,n}$ be the mixture posterior mean after the signals, and let $A_n := \mathrm{Var}_{\mathbf{w}_n}(m_{\cdot,n})/(2\lambda)$ be the hedging cost the agent faces at step (4), the analogue of (11) at the updated weights and means.

**Proposition 4** (The value of information with updating)**.** *Under* (8) *with* $\varphi$ *linear and the update* (15)*, the ex-ante value to the agent of the n signals, evaluated under the mixture* $\mathbf{w}$*, is*

$$\mathrm{VOI}(\mathbf{w},n) \;=\; \frac{\mathrm{Var}_{\mathbf{w}}(\bar{m}_n)}{2\lambda} \;=\; \underbrace{\frac{\sum_{k=0}^K w_k\,\psi_k(n)}{2\lambda}}_{\textit{within-model refinement}} \;+\; \underbrace{A(\mathbf{w})-\mathbb{E}_{\mathbf{w}}[A_n]}_{\textit{resolution of model uncertainty}}. \tag{16}$$

*(a)* $\mathrm{VOI}(\mathbf{w},n)\geq 0$ *for every* $(\mathbf{w},n)$*: the agent is never averse to information.*

*(b) The resolution component vanishes at every vertex of the simplex and converges to the hedging cost,* $A(\mathbf{w})-\mathbb{E}_{\mathbf{w}}[A_n]\to A(\mathbf{w})$ *as* $n\to\infty$*: in the limit, information recovers exactly the cost of holding several mental models at once.*

*(c) Under Assumption 1,* $\mathrm{VOI}(0,n)=0<\mathrm{VOI}(\mathbf{w},n)$ *for every* $\mathbf{w}\neq 0$ *and* $n\geq 1$*: curiosity is absent under the default alone and present once an alternative is activated.*

*(d) The doubt is resolved in one of two ways. Take a single alternative k held against the default. If both mental models read the signals as informative about* $\theta$*, then* $m_{k,n}-m_{0,n}\to 0$ *while* $w_{k,n}$ *converges almost surely to a limit strictly between* 0 *and* 1*: the models come to recommend the same action, so the hedging cost disappears, but the question of which model is right is never settled. Under Assumption 1,* $w_{k,n}\to 1$ *almost surely if the signals are generated as the alternative says and* $w_{k,n}\to 0$ *if they are generated as the default says: the question is settled and the agent keeps one mental model.*

Part (b) makes precise what the introduction calls resolving the model uncertainty that activation creates: the hedging cost is by construction the value of learning which mental model is right, and enough signals deliver that value in full. Part (d) separates two things "resolving model uncertainty" can mean: the models coming to agree, or one of them winning. The design principles of Section 4.1 turn on the difference. Part (a) is a statement about a linear $\varphi$; under a concave $\varphi$ the ex-ante value of information is not signed in general (Li 2020), and the results on the weights chosen at step (2) do not depend on its sign.

### 3.5 What moves the evidence bound: context effects

In psychology, a context effect is the finding that cues present when an experience is stored and when it is recalled shape what is remembered, even when the cues are irrelevant to the task (Tulving and Thomson 1973; Godden and Baddeley 1975; Smith and Vela 2001); the temporal context model formalizes how context drives retrieval (Howard and Kahana 2002; Kahana 2012). Economic models of associative memory build on this finding: a payoff-irrelevant cue changes which experiences or representations come to mind, and so changes beliefs and choices (Gennaioli and Shleifer 2010; Bordalo et al. 2020; Enke et al. 2024; Bordalo et al. 2026).

Propositions 1 and 3 take the evidence bounds as given. This subsection opens them up. Because (1) is a linear Gaussian model, each $L_k$ is available in closed form and each Bayes factor decomposes into three interpretable channels, which is what turns the likelihood condition from a constraint into a design instrument. Under (1), $x \sim \mathcal{N}(\mu_k b_k,\ \Sigma + \sigma_k^2 b_k b_k')$ under mental model $k$, and $L_k(x)$ is that density evaluated at $x$ (Lemma A.4 in the Appendix).

**Proposition 5** (Three-channel decomposition of the evidence bound)**.** *Define, for each mental model $k = 0, 1, \ldots, K$,*

$$\underbrace{R_k := x'\Sigma^{-1}x - s_k^2/v_k^2}_{\textit{off-attention residual}}, \qquad \underbrace{q_k := \frac{(s_k-\mu_k)^2}{\sigma_k^2+v_k^2}}_{\textit{standardized surprise}}, \qquad \underbrace{c_k := \log\left(1+\frac{\sigma_k^2}{v_k^2}\right)}_{\textit{Occam factor}}.$$

*Then $R_k \geq 0$ and, for each alternative $k$,*

$$\log \Lambda_k(x) \;=\; \tfrac{1}{2}\Big[\underbrace{(R_0 - R_k)}_{\textit{encoding}} + \underbrace{(q_0 - q_k)}_{\textit{surprise}} + \underbrace{(c_0 - c_k)}_{\textit{Occam}}\Big]. \tag{17}$$

*Consequently:* (i) *the encoding channel vanishes identically if and only if $b_0 \propto b_k$;* (ii) *the Occam channel is strictly negative whenever $\sigma_k^2/v_k^2 > \sigma_0^2/v_0^2$, in particular whenever the alternative has the default's loading, $b_k = b_0$, and the more diffuse prior; and* (iii) *$\overline{W}_k(x)$ is increasing in each of the three channel differences of alternative $k$, decreasing in each channel difference of every other alternative, and equals the Bayesian posterior model probability.*

Two properties follow. First, the default can become indefensible by two distinct routes: the observation is extreme on a signal both models read (the surprise channel $q_0 - q_k$), or it loads on directions the default does not read at all (the encoding channel $R_0 - R_k$). The second route is moved by which features of the same observation are read first, with its objective content held fixed. It also determines when context matters. Call a component $z$ of the observation *payoff-irrelevant under the default* if the default does not read it, $b_0'\Sigma^{-1}z = 0$.

**Corollary 2** (Context effects)**.** *Let $x$ and $x+z$ be two observations, where $z$ is payoff-irrelevant under the default.* (i) *If no mental model on the menu reads $z$, that is, $b_k'\Sigma^{-1}z = 0$ for every $k$, then $\Lambda_k(x+z) = \Lambda_k(x)$ for every $k$, so the evidence bounds, the weights, the information acquired, and the act are all unchanged.* (ii) *Otherwise $z$ moves $\Lambda_k$ only for the alternatives $k$ that read it, through their statistics $s_k$, and moves the bound of any other alternative only through the denominator of* (5)*.*

The proof is immediate from (17): $x'\Sigma^{-1}x$ cancels from $R_0 - R_k = s_k^2/v_k^2 - s_0^2/v_0^2$, so every term of $\log\Lambda_k$ depends on $x$ only through the statistics $s_j = b_j'\Sigma^{-1}x/(b_j'\Sigma^{-1}b_j)$, and a component no model reads leaves

each of them unchanged. In memory-based accounts a payoff-irrelevant cue changes which representation is retrieved and so changes behavior (Bordalo et al. 2026). Here it sways decisions only when another model makes it relevant, it must still clear the activation threshold of Proposition 1, and if it does, the search that follows tests the shift rather than entrenching it (Corollary 4(i)).

The corollary separates two questions that memory-based accounts answer together. Whether context moves behavior depends on the menu of mental models: some model must read the cue. Whether the movement lasts depends on the information environment. When evidence that discriminates between the models is accessible and generated by the truth, a mistaken shift prompted by context is transient, whatever direction the agent chooses to look in (Remark 1). A context effect that persists therefore points to evidence that is scarce, too costly to seek given the weight the alternative carries (Corollary 1), or biased toward the mistaken model.

Second, every alternative begins at a handicap. A marginal likelihood penalizes a model for spreading predictive mass, and $c_k$ is the Occam factor of Bayesian model comparison (MacKay 1992; Jefferys and Berger 1992; Kass and Raftery 1995; Ba 2026). Since $c_k$ is increasing in $\sigma_k^2$, an optimistic but vague alternative ("the program might really help") is *harder* to justify than a specific one ("pass rates rise by about eight points"), even when the vague version has the higher mean. Wishfulness does not buy likelihood; precision does.

### 3.6 Interference: divided evidence activates none

In psychology, interference is the finding that similar memories compete at retrieval, so that recalling one item makes related items harder to recall (Underwood 1957; Anderson and Neely 1996; Anderson 2003; Kahana 2012). In economic models of associative memory, experiences associated with other hypotheses interfere with the retrieval of those that bear on the hypothesis being assessed, which distorts probability judgments (Bordalo et al. 2023).

The menu delivers a result that no comparison of one alternative with the default can. When each of several alternatives is worth more than the default, none may be activated, because the same evidence must justify each of them and justifies none enough: the evidence bound of each alternative is deflated by every other alternative that looks plausible, the agent's attention is rivalrous across them, and the superadditive cost of holding several priors at once raises the price of search. Two forces make a menu of alternatives harder to act on than a single one: the cost function penalizes spreading attention (Lemma A.2), and the evidence bounds crowd one another. This is interference, but not the kind memory models study, where a competing cue changes which representation is retrieved. Here every alternative remains available and none is forgotten; rival explanations interfere by diluting one another's support.

**Proposition 6** (Fragmentation)**.** *Fix $x$ and let $\overline{W}_k$ be given by* (5)*.*

*(i)* (Crowding.) *For each $k \geq 1$, $\overline{W}_k$ is strictly decreasing in $L_j$ for every $j \neq k$ with $j \geq 1$. In particular $\overline{W}_k < L_k/(L_0 + L_k)$, the bound that would obtain were $k$ the only alternative on the menu.*

*(ii)* (Breadth dilutes.) *Let a treatment scale the likelihoods of a set $S \subseteq \{1,\ldots,K\}$ by a common factor $\rho > 1$. For $k \in S$, the resulting bound is strictly smaller when $|S| \geq 2$ than when $S = \{k\}$.*

*(iii)* (Joint failure.) *Consequently, in the all-or-nothing regime, a feature vector that raises several $L_k$ moderately can leave every $\overline{W}_k$ below the level alternative $k$ needs in Proposition 1, in which case $w_k^* = 0$ for all $k \geq 1$ and, under Assumption 1, $n^* = 0$. In the graded regime crowding caps each weight at $\overline{W}_k$, and no information is acquired once every $\overline{W}_k$ falls below the level in Corollary 1.*

The signal's diagnostic power, large in aggregate, is fragmented across the menu, and the default remains defensible. Where crowding lowers the bounds without pushing them below threshold, alternatives are activated at smaller weights; since information about an alternative is acquired only once its own weight clears a level (Proposition 8(i)), search is weaker and the shift slower, so the joint failure of part (iii) is the limit of a graded effect rather than a knife edge. The property has an exact form. Model an informational treatment as a

budget $B \geq 0$ allocated across alternatives, $\beta = (\beta_1, \ldots, \beta_K)$ with $\beta_k \geq 0$ and $\sum_k \beta_k = B$, which multiplies $L_k$ by $e^{\beta_k}$. This normalization holds fixed the total log-likelihood content delivered and varies only its concentration.

**Proposition 7** (The ceiling on untargeted information)**.** *Let $\overline{W}_k(\beta)$ denote the post-treatment bound* (5)*.*

*(i)* (Targeting.) *Order the alternatives so that $L_1 = \max_k L_k$. For fixed $B$, $\max_k \overline{W}_k(\beta)$ is maximized at the vertex $\beta_1 = B$. Along that allocation $\overline{W}_1$ is strictly increasing in $B$ and $\overline{W}_1(B) \to 1$ as $B \to \infty$.*

*(ii)* (Breadth.) *Under the uniform allocation $\beta_k = B/K$, $\max_k \overline{W}_k$ is strictly increasing in $B$ but*

$$\sup_{B \geq 0} \max_k \overline{W}_k(B) \;=\; \frac{\max_k L_k}{\sum_{j=1}^K L_j} \;<\; 1.$$

*(iii)* (Non-comparability.) *Consequently, for any activation threshold $T \in \left(\max_k L_k / \sum_j L_j,\ 1\right)$, a targeted treatment activates some alternative for $B$ large enough, while a uniform treatment fails to activate any alternative at* every *budget, however large.*

Breadth fails not because it delivers less information, since by construction it delivers the same, but because spreading it raises every likelihood in the denominator of every bound, so the ratios barely move. Panel C of Figure 2 shows the two paths for $K = 4$: the targeted bound passes 0.9 at a budget of about four, while the uniform bound is still below 0.29 at any budget whatsoever.

The ceiling itself is a property of normalized posteriors, and Bayesian model averaging combined with any fixed activation threshold would share it. What the model adds lies elsewhere: the threshold $T_k(\overline{W}_k)$ is derived rather than set, and rises with the disagreement $D_k$ between the actions the models recommend; crowding is weighted by fit, since a rival enters the denominator of (5) through its likelihood, so one the observation does not support barely crowds, unlike attention drawn by salience; and below the threshold the weight, and with it curiosity, is exactly zero rather than merely small, as it would be under averaging. Corollary 3 makes the consequence precise: curiosity is non-monotone in the number of rivals, and in the all-or-nothing regime it reaches zero after finitely many.

**Corollary 3** (Curiosity and the number of rivals)**.** *Fix $x$ and let the menu contain $K$ alternatives that are symmetric given $x$, each with likelihood $L$, value gap $\Delta U$, and disagreement $D > 0$, so that each bound is $\overline{W}(K) = L/(L_0 + KL)$, strictly decreasing in $K$ and tending to zero. Curiosity $A$ and information acquired $n^*$ are zero at $K = 0$; positive at $K = 1$ whenever the alternative is activated at a weight above the level in Corollary 1; and tend to zero as $K$ grows. Moreover $n^* = 0$ for every $K$ beyond a finite bound, and so is $A$ in the all-or-nothing regime when $\Delta U < D$.*

By Lemma A.2 at most one alternative is held, at a weight $w \leq \overline{W}(K)$, so $A = w(1-w)D \leq \overline{W}(K)D$ by (12); information is acquired only once $w$ exceeds the level in Corollary 1, which $\overline{W}(K)$ eventually falls below; and in the all-or-nothing regime the threshold $T(\overline{W}) = D(1-\overline{W}) + \frac{\gamma}{2}\overline{W}$ rises toward $D$ as $\overline{W}$ falls, so activation fails once $\overline{W}(K) < (D - \Delta U)/(D - \gamma/2)$.

**Stacking.** Finally, suppose two alternatives must eventually be entertained. Under (3) the cost of holding both at weights $w_1, w_2$ is $\frac{\gamma}{2}(w_1^2 + w_2^2) + \eta w_1 w_2$, whereas two separate activation–signal–resolution cycles cost $\frac{\gamma}{2}(w_1^2 + w_2^2)$ in total: holding them simultaneously costs a stacking premium of exactly $\eta w_1 w_2$, and a larger hedging cost whenever the two models imply different posterior means, so absent discounting sequencing weakly dominates stacking (Proposition A.1 in the Appendix).

### 3.7 Endogenous exposure: what the agent looks at

So far the agent chooses how much information to acquire but not what to look at. Let her choose signal counts $(n_1, \ldots, n_K)$, one per alternative, with type-$k$ signals loading on features only model $k$ encodes, so that

Assumption 1 holds type by type; and let her choose the direction of the next observation, $y = d'\tilde{x} + \eta$ with $\tilde{x}$ a fresh draw of the features and $\eta \sim \mathcal{N}(0, \nu_y^2)$. A school visit that looks at crop yields is one $d$; one that looks at what students can do is another. Write $\delta := b_k m_k - b_0 m_0$ for the difference in the two models' predictions and $\Omega_0 := \Sigma + b_0 b_0'/P_0(0)$ for the default's predictive covariance.

**Proposition 8** (Endogenous exposure)**.** (i) Divided attention buys nothing. *With type-specific signals the information problem separates by alternative: for each k, $n_k^*(w_k) = \nu_y^2[(w_k/(2\lambda\nu_y^2 c))^{1/2} - P_k(0)]^+$, so no information is acquired about alternative k unless $w_k$ alone exceeds $2\lambda\nu_y^2 c\, P_k(0)^2$. Total attention off the default is not sufficient for any information to be acquired: the same $W = \sum_k w_k$ clears the threshold when concentrated on one alternative and clears none when divided.* (ii) Curiosity is directed at disagreement. *With a single alternative k held against the default, among directions normalized by $d'\Omega_0 d = 1$ the one that maximizes the squared difference between the two models' predictive means of y is*

$$d^* \;\propto\; \Omega_0^{-1}\big(b_k m_k - b_0 m_0\big),$$

*with value $\delta'\Omega_0^{-1}\delta$. It is the direction along which the two models disagree most about what will be seen, and it differs from the alternative's own attention mapping $\alpha_k$ generically when $b_k \not\propto b_0$.*

Part (i) is the demand-side counterpart of fragmentation: divided evidence activates no alternative, and divided attention investigates none, so the "held but not investigated" path is the generic outcome of a spread of small weights. Part (ii) says what an activated agent looks at. One who valued only refinement within her favored model would look along that model's attention mapping; one who values resolving which model is right looks where they disagree, a different direction exactly when the two are distinct mental models in the sense of Section 3.5. This is a formal version of the information-gap account of curiosity (Loewenstein 1994): the gap is the predictive disagreement between the models held.

**Corollary 4** (Exposure and the path of the shift)**.** (i) *If observations along $d^*$ are generated as alternative k says, each raises* $\log\Lambda_k$ *in expectation by the Kullback–Leibler divergence between the two predictive distributions, which is increasing in $\delta'\Omega_0^{-1}\delta$: the shift continues on the agent's own exposure. If they are generated as the default says, each lowers* $\log\Lambda_k$ *in expectation and the agent returns to the default. At $w_k = 0$ no observation is sought and neither occurs.* (ii) *If several alternatives are plausible after x and one clears its threshold, it is investigated first; if the evidence goes against it, its likelihood falls and by Proposition 6(i) every other alternative's bound rises, so a second alternative may clear its threshold in turn. If no alternative clears its threshold after x, no investigation starts, no likelihood changes, and no bound moves: the joint failure of Proposition 6(iii) is absorbing until a new exogenous observation arrives.*

Part (i) delivers persistence from choices rather than from a state variable: a shift that crosses the threshold carries itself forward when the alternative is right and corrects itself when it is wrong, because the agent now looks where the answer lies, while below the threshold her exposure stays governed by the default and nothing she sees moves the bound. Part (ii) sharpens targeting: she can climb out of a fragmented menu on her own, eliminating hypotheses one at a time, but only after something has lifted one alternative over its threshold.

*Remark* 1 (Misdirected search). Self-correction in Corollary 4(i) does not require the agent to look along $d^*$. For any direction $d$, an observation generated by the true model changes the log Bayes factor of a wrong alternative, in expectation, by minus the Kullback–Leibler divergence between the two models' predictive distributions along $d$, which is nonpositive and strictly negative whenever those distributions differ; and by Lemma A.3(i) the odds of the alternative against the default cannot exceed its updated Bayes factor. A distortion of where the agent looks, such as a tendency to seek evidence that would confirm the alternative, therefore slows correction, and stalls it only along directions on which the models make identical predictions, but cannot reverse its direction. Entrenching a mistaken shift requires evidence that is itself biased toward it.

**Ambiguity aversion and the direction of curiosity.** The closed forms set $\varphi$ linear, and Remark A.1 records that concavity raises the activation threshold. It also changes what the agent is curious about, and this is where the smooth ambiguity aggregator does its own work. Holding alternative $k$ at weight $w_k$ carries an *ambiguity premium* $\frac{\rho}{2}\varphi'(\bar{U})\, w_k(1-w_k)(\Delta U_k)^2$ beyond the hedging cost, which a signal changes through the weight and the value gap it leaves behind.

Under the second-order expansion, the value of signals to an ambiguity-averse agent equals its value under linear $\varphi$ plus a term that is nonnegative for signals which move only the weight, discriminating between the models without refining either, and nonpositive for signals which only refine them, the first because the weight is a martingale and $w(1-w)$ concave, the second because the value gap is spread without shifting its mean (Proposition A.2 in the Appendix). An ambiguity-averse agent is thus more curious about the features on which the models disagree (Proposition 8(ii)) and less curious about outcome data both models can absorb, than a linear agent with the same beliefs would be. The smooth ambiguity aggregator, which plays no role in the closed forms, is what sets the *direction* of curiosity: the more the agent dislikes model uncertainty, the more her information seeking is aimed at ending it rather than at refining the model she favors.

### 3.8 The neuroscientific regularities the model reproduces

The paper's claim is that a well-established neuroscientific theory disciplines the decision problem, and that claim is testable against the theory's own findings. Table 1 in the Appendix matches nine documented regularities to the model feature that produces each. Three correspondences are close. The evidence bound is, term for term, the posterior over contexts that Sanders et al. (2020) argue governs remapping, so the model's central constraint is not an analogy but the same object placed in a decision problem. The two regimes of Proposition 1 reproduce rate and global remapping (Leutgeb et al. 2005a,b; Wills et al. 2005), and the model supplies what the neuroscience does not have, a classifying variable: the disagreement between models about what to do relative to the cost of effort. And the interference term reproduces the working-memory capacity limit (Cowan 2001; Bays and Husain 2008) in the only form a decision model can use, a superadditive cost.

Two are looser. Flickering between maps (Jezek et al. 2011) is a within-trial phenomenon on a timescale of seconds, and the model's interior weights are its natural counterpart but have no timescale; and global remapping replaces one code by another, whereas the abrupt regime moves the weight to the evidentiary limit $\overline{W}_k < 1$. On three matters the model is silent: the geometry of the code, the automatic assignment itself, which it takes as given in (6), and whether the variable that sorts an animal into rate or global remapping is the model's $\gamma$ against $2D_k$. The last is the honest gap, since those experiments manipulate environmental similarity rather than disagreement about a payoff.

### 3.9 What distinguishes the model

Table 2 in the Appendix lists the phenomena delivered above and marks which of the closest alternatives also deliver them: Bayesian model averaging, hypothesis testing in the sense of Ortoleva (2012), model persuasion in the sense of Schwartzstein and Sunderam (2021), optimal expectations (Brunnermeier and Parker 2005), and associative-memory categorization in the sense of Bordalo et al. (2026). Two rows carry most of the weight. The one-sidedness of the constraint is what separates the model from Bayesian model averaging: a Bayesian is moved by evidence exactly as much as Bayes' rule says, whereas the agent here may stay with the default against the evidence but cannot run ahead of it, which is why the default is sticky and why interventions must clear a threshold. And the activation threshold is what separates it from hypothesis testing and model persuasion: those frameworks have the agent switch when the evidence is decisive, but they have no counterpart to holding an alternative at a small weight, investigating it or not, and being crowded by other alternatives, which is where the model's design content lives.

**Behavioral puzzles.** Several regularities in belief updating and information acquisition, usually explained separately, follow from the same structure. Underreaction to evidence (Edwards 1968; Benjamin 2019) is the one-sidedness of the constraint: below the activation threshold the weight stays at zero although the Bayesian bound is positive. That underreaction coexists with abrupt revisions is the all-or-nothing regime of Proposition 1, in which the same agent ignores weak evidence and jumps on strong evidence. The avoidance of freely available, decision-relevant information (Golman et al. 2017) needs no taste for ignorance: below the threshold there is no second model on which the information would bear, so it has no value (Corollary 1). The failure to notice features that matter (Schwartzstein 2014; Hanna et al. 2014) is the attention mapping of the model in use (Lemma 1), which changes, and with it what is noticed, only when weight moves. And curiosity that peaks at intermediate confidence (Loewenstein 1994) is the inverted U of Definition 1 in the weight.

# 4 Application to Information Interventions

The model applies to the design of information interventions in a program of school-based agricultural education (SBAE) in Liberia (Lee 2025). SBAE offers to break the vicious cycle in which low agricultural productivity and household labor needs depress school attendance. Its theory of change rests on two channels of technology diffusion to rural households: (i) school farms that demonstrate improved techniques to parents, and (ii) student home projects that test the appropriateness of those techniques on the household's own soils. The information interventions studied here aim to draw parents' attention to the program; they are problems of activating a fixed alternative: the "SBAE" model of agribusiness skills and education prospects against the default "school garden" model of drudgery and students as farm assistants. What an intervention must move is the likelihood ratio between the two, through contrast. The default is, in addition to being more precise than the alternative, pessimistic about the program; that is a feature of the application rather than of the model. The design principles come first and the application follows.

## 4.1 Design principles

The properties of Section 3 translate into five design principles for informational interventions in settings where stimulating continuous model exploration is crucial, each illustrated with the SBAE program.

**Build contrast from a shared foundation.** An intervention need not diagnose each agent's default. It can start from a mental model the population holds in common, show what that model cannot answer, and then introduce one alternative that does, so that the default is made indefensible against that alternative. Activation requires the value and evidence conditions together (Proposition 1), and interventions typically fail on the second: facts the default can absorb leave every bound near zero. By Proposition 5 the bound can be raised on three margins (Table 1): surprise after projection through the default's attention mapping; features the default does not read, made prominent; and a specific rather than a vague alternative. A promotional video opening with students' agribusiness skills rather than garden labor works on the second; a narrow falsifiable claim in place of a broad encouraging one works on the third. Once the threshold is crossed the loop of Section 3.3 starts on its own.

**Provide discriminating observations.** By Proposition 8(ii) an activated agent looks where the models disagree about what will be seen, not where the alternative alone is informative. An intervention should therefore make discriminating observations available rather than merely informative ones: a school visit showing crop yields is read by both models and moves the weight little, while one showing what students can do, skills the "school garden" model predicts should not be there, is discriminating. By Corollary 4(i) her own exposure then carries the shift forward if the alternative is right and back if it is wrong, so the intervention need only

Table 1: Intervention margins

| Margin | What it changes | Example, objective content held fixed |
|---|---|---|
| Activation ($w_k$) | Raises $\Delta U_k$ or the surprise channel $q_0 - q_k$. | Credible results that raise $U_k$; evidence that is surprising after projection through $\alpha_0$. |
| Encoding ($\alpha$) | Changes the residual channel $R_0 - R_k$: which features of the same raw $x$ are read. | An opening sentence or shot order that leads with skills rather than garden labor. |
| Specificity ($\sigma_k$) | Shrinks the Occam penalty $c_k$. | A narrow falsifiable claim in place of a broad encouraging one. |

have crossed the threshold and left the discriminating observations within reach. This is distinct from contrast, which acts on the observation the intervention controls.

**Supply alternative-specific content, not outcomes.** By Proposition 4(d), outcome data on a scale both models read make them agree about what to do without deciding between them, leaving the weight parked wherever the evidence left it. Only content for which the default has no slot, the mechanism connecting a school farm to a marketable skill, moves the weight to a corner, which is also what Assumption 1 presumes.

**Target rather than spread.** Holding aggregate informational content fixed, a treatment that concentrates content on a single alternative activates it at budgets for which a treatment distributing the same content across several alternatives activates none, and the failure of breadth has a ceiling that no budget overcomes (Propositions 6 and 7). Enumerating many reasons a program might work makes several alternatives moderately plausible and none plausible enough; one integrated case is what clears a bound.

**Sequence rather than stack.** When several alternatives must eventually be entertained, delivering them one at a time, letting one activation–signal–resolution cycle complete before the next begins, saves the stacking premium $\eta w_1 w_2$ and the extra hedging cost (Proposition A.1 in the Appendix). Delivering the alternatives in separate sessions rather than in a single block is an instance. Together the five principles say: narrow, deep, and discriminating. The first says make the default indefensible against one alternative; the second says leave the observations that would settle the matter within reach; the fourth and fifth add, do not make several alternatives look equally plausible at the same time.

## 4.2 Parents: why was parental engagement pivotal?

Lee (2025) shows that parental engagement (an informational component centered on a promotional video that clarifies how SBAE differs from typical school gardens, which merely employ students as farm assistants) is what activates SBAE's two channels. With it, SBAE is more cost-effective than most African extension programs in diffusing technology and twice as cost-effective as conditional cash transfers in raising attendance. Without it, SBAE fails to improve adoption on any household parcel and *reduces* students' studying hours and attendance.

The reversal points to an aspirations failure rather than an information friction. A large literature studies "pure" information frictions, in which objective facts exist but beliefs deviate from them because the facts are inaccessible (Jensen 2010), costly to obtain (Bergman 2021), or hard to process (Dizon-Ross 2019). But beyond a lack of facts, participants may misclassify a program into a category that demotivates them. Reports from across Africa (FAO 2009) and my own qualitative fieldwork suggest that parents commonly associate school garden programs with drudgery that trains students as obedient farm assistants rather than equipping them with agribusiness skills. Because that preconception is widely shared, the intervention need

not diagnose individual parents: it builds contrast against a model the population holds in common.[9] More puzzling is parents' persistent avoidance of information that would resolve the uncertainty: in three years, with 81% student participation, only 44% of parents visited school farms, and absent the informational treatment SBAE-school parents *reduced* visits and parent–teacher meeting attendance by 39% relative to control.

These patterns are hard to rationalize with a fixed prior. Visits are costly but not prohibitively so, so under diffuse priors their informational value should exceed the cost; a reduction indicates priors concentrated enough that observing SBAE is read as confirmatory. An imprecise negative signal does not account for it either: under a single prior a visit's value depends on how uncertain the parent remains rather than on how bad the news was, so such a signal raises the value of visiting. And the only signal SBAE-only parents receive, their children's report that school time is spent on a farm, is bad news under one reading of the program and good news under another; its sign is supplied by the mental model, not by the message.

The serious version of the alternative adds a fixed cost of visiting, which creates a region of beliefs in which information is not worth acquiring. A sufficiently precise negative signal moves the parent out of that region and she stops checking; the informational treatment moves her back in; and her own subsequent learning does the rest. On this reading the video need not carry the weight of the conclusion, so neither the size of the reversal nor its persistence is evidence against updating. Three features of the setting nonetheless sit awkwardly with it. First, the comparison is with parents at schools that received nothing, so the account requires them to conclude that a school given a farm, training and materials is less worth engaging with than one given none. Second, both arms face the same program and the same accessible evidence, and in the engaged arm that evidence is favorable enough to sustain the reversal; the signal that supposedly drove the other arm out of engagement must then have been not merely unfavorable but misleading, and misleading in the same direction for enough parents to move the average. Confident error of that kind is a property of the reading, not of the draw. Third, the response extends to study hours and school attendance, whereas a signal about one program's quality should move engagement with that program.

Two qualifications are in order. Returns may be complementary in parental engagement, in which case the unfavorable reading is correct rather than misleading and the treatment coordinates parents onto a better outcome; that is a distinct hypothesis, and one that implies effects should depend on how many other parents at a school were treated, which the framework here does not require. And no reduced-form comparison can exclude the class of explanations that work through the level of beliefs, since any response admits some unobserved content that would rationalize it. What separates the accounts is variation that holds content fixed, which is the first of the predictions below. Selective attention (Schwartzstein 2014; Hanna et al. 2014) would imply parents are unaware of science instruction, new techniques, or home projects, whereas anecdotally they are aware of these but lack the mental model that makes agriculture and education complements. And associative memory (Bordalo et al. 2020, 2022) explains why cues retrieve unfavorable episodes, but not why parents fail to seek information: a parent retrieving poor memories can still recognize that her simulation is uncertain.

**An explanation based on endogenous model uncertainty.** Endogenous model uncertainty offers a new lens into aspirations failures: the failure to aspire is a failure to activate new mental models. The parent trades off attention between a costless, familiar model ("failed school gardens") offering a precise but limited prospect and a costly new model ("an innovative program offering new techniques, practical skills, and better education prospects") that, while less precise, offers higher potential utility. This generates a threshold effect: the signal must render the familiar model indefensible to justify the cost of shifting attention (Proposition 1). Once the new model carries positive weight, model uncertainty arises endogenously and drives the parent

[9] Ashraf et al. (2025) present randomized-trial evidence hinting at similar aspirations failures: in regions of Colombia where government interventions have limited effectiveness, an entrepreneurial training program reduces low-income, traumatized populations' ability to imagine a positive future through business and worsens economic outcomes, whereas adding a mental intervention that helps participants simulate pathways between actions and outcomes removes the negative effects.

Table 2: Applications to pedagogy and learning and the model objects they map onto

| Agent | Default mental model | Alternative(s) | What an intervention must move |
|---|---|---|---|
| Teachers | "Banking": cover the syllabus, drill for the exam | "Inquiry": pose questions, frame hypotheses, reason from evidence | Weighing the new model against the old (reflection); sharpening the new model (experience with peers) |
| Students | "School": the written step-by-step procedure | "Work"; "integrated" (school and work skills as one domain) | Building the integrated model; avoiding rivalrous attention across models |
| Farmers | "They simply had a good year" | Competing hypotheses about why yield rose | Cognitive dimension reduction: why a technique works and under what conditions |

*Notes.* In every row the default mental model is familiar and precise, with a low mean; the alternatives have higher means and lower precision.

to seek further signals (Propositions 3 and 4), and once she chooses what to look at the loop sustains itself (Corollary 4). The role of an informational treatment is thus not to trigger updating under a fixed prior but to disrupt the preconception that stifles acquisition.

Three predictions distinguish this from standard learning models. First, contrast moderates treatment effects through the likelihood ratio between the models: presenting a video around the new model ("this is an innovative, successful program, with the following benefits…") can have the opposite effect of presenting it around the preconception of familiar school gardens.[10] Second, model uncertainty arises endogenously when the treatment disrupts the preconception: parents become less confident in the familiar model, a moment of metacognition, begin to assign weight to the new one, and hold more ambiguous assessments of concrete program events. Third, information-acquisition effort diverges over time across the two presentations, so they produce only small behavioral differences in the early stages.

# 5 Application to Pedagogy and Learning

The remaining interventions of the SBAE program aim to promote new teaching, learning, and farming approaches by fostering a conceptual understanding of pedagogy choice and learning efforts among teachers, students, and farmers. Each is again a problem of activation: a familiar and precise default governs behavior, one or more alternatives are worth more but less precise, and what an intervention must move is the evidence that decides between them. The three settings differ in what makes activation hard: the teacher has never seen the alternative, the student holds the right models but never brings them together, and the farmer faces too many at once. The same five design principles apply to each (Table 2).

## 5.1 Teachers: reflection and experience are crucial for continuous shifts in pedagogy

Integrating experiential elements into science education is constrained by the fact that many teachers have never experienced that form of education. Rural education in Liberia devotes much of its time to rote memorization over conceptual understanding, and many junior high science teachers lack formal training in pedagogy, so program officers report that teachers recite recipes from training, explaining what a soil profile is on the school farm, without adapting to their own curriculum. Studying a teacher-skills training approach called "Learning to Learn" (LtL), Nourani et al. (2025) provide experimental evidence that this can be overcome with an immersive five- to six-week training blending hands-on pedagogical exercises with reflective study,

[10]The "school garden" presentation reinforces the existing template and triggers filling-in from prior experience, called pattern completion in neuroscience; the "SBAE" presentation triggers encoding of the program as distinct from superficially similar experiences, called pattern separation. In the model these are the encoding and Occam channels of Proposition 5.

which seeks to transform how teachers think about knowledge and learning rather than to supply recipes. The decision is whether the teacher activates a new mental model of what the classroom can be; he is forming beliefs about the returns to experiential strategies, and those beliefs vary with the model he applies.

The default is the "banking" model (Freire 1970): cover the syllabus, write on the board, drill for the exam. Applying an experiential strategy under it means following a recipe, whose outcome is certain but whose effect is limited, because the experiential element is a visual extension of existing material. The alternative is the "inquiry" model LtL promotes: pose sharp questions, frame hypotheses, reason from everyday evidence, and help students find answers rather than supply them. Its returns are potentially high, since the visual aid becomes a cue that structures inquiry, but a teacher who has never experienced it is highly uncertain about them. Conventional workshops, including "SBAE only," fail because they supply strategies without the stimulus to keep exploring their returns: the workshop provides a positive signal, but back in the classroom the teacher reads it under the default, never activates the alternative, and stops acquiring signals.

LtL supplies two design features that conventional training lacks. First, after working through modules on real-world topics, teachers attend reflection sessions that confront how poorly the traditional model accounts for the learning they want. In the model, reflection works on the evidence condition, raising the evidence bound of the "inquiry" model through the surprise and encoding channels of Proposition 5, the first design principle. Second, teachers experience the materials with peers, building a mental image of how an inquisitive classroom feels, which sharpens the precision of the "inquiry" model, lowers its Occam penalty, and raises the value of further signals under it. Together these sustain activation and acquisition after the training ends, which is the complementarity of Proposition 3; and because sessions are spread over weeks, cycles run one at a time, as the fifth design principle recommends.

### 5.2 Students: fragmented skills need to be integrated

SBAE's theory of change is that students come to recognize the usefulness of academic knowledge through its connections to agriculture. But not all scientific knowledge bears on agriculture, and more fundamentally students often fail to recognize work-related knowledge as belonging to the same domain as what is taught in school: Banerjee et al. (2025) find that children who perform fast, accurate mental arithmetic at market stalls stumble on the same arithmetic on a school worksheet, while schooled children show the reverse. The decision is whether a working student activates a relevant mental model for the academic problem in front of him.

An academic problem triggers the "school" model by default, the written, step-by-step procedure, whose returns he understands but views pessimistically, not having mastered the formal procedures. Two alternatives exist. Under the "work" model he has accumulated relevant skills: measuring crop spacing, identifying parts of a plant, running comparisons across plots, calculating quickly at farm sales. It favors rapid shortcuts over formal steps, and its returns to studying are less certain but real. Under the "integrated" model he recognizes that work shortcuts and school procedures are both legitimate problem-solving skills that correspond to one another; he is highly uncertain about its returns, but they exceed the "work" model's, whose shortcuts apply to only a subset of academic problems. SBAE's experiential component raises precision about the "work" model and lengthens study hours (Lee 2025), but its effects are mixed, because the shortcuts fit only some problems, so a mix of models is activated, and holding several at once strains attention (Section 3.1, Proposition 6).

LtL leads students to appreciate conceptual understanding and to pair it with the precise use of language, the formulation of hypotheses, and evidence drawn from everyday life, which sharpens the precision of the "integrated" model; and because the trained teacher is given no recipes, he directs experiential learning toward curriculum needs.[11] LtL thus works on both margins the model identifies: it presents exactly the two ways of solving a problem that are to be seen as one, so a single alternative is singled out rather than several made

[11] One previously trained teacher remarked: "I have been teaching for 30 years and am only now realizing that all I was ever doing was helping my pupils memorize information."

plausible (Propositions 6 and 7), and it sharpens that alternative, which raises its evidence bound through the Occam channel (Proposition 5) and the value of further exploration under it (Proposition 3).

### 5.3 Farmers: high-dimensional learning and the case for scientific understanding

African agricultural technology diffusion is constrained by a lack of extension officers, which social learning can partly overcome. But farmers learn from messy observation rather than controlled experiment, and the suitability of techniques varies sharply across plots and households (Conley and Udry 2010; Suri and Udry 2022). Consider a farmer who observes a surprising yield on a neighbor's plot using SBAE-promoted techniques and can entertain many hypotheses about why. Every mental model here is such a hypothesis; the decision is whether he activates one beyond the default and which, and he is forming beliefs about the returns to experimentation. The distinguishing feature is high dimensionality, since the experiment was uncontrolled.

Because the neighbor applies many traditional techniques alongside the new one, the plot reads as farmed in the ordinary way, and the default is "they simply had a good year," precise from accumulated experience and low in mean. Consider four alternatives: the intercropped legume fed the cassava, so the benefit is tied to that crop combination; the raised mounds drained the heavy clay, so it is specific to poorly draining soil; the flat lowland held moisture, so it depends on a topography the household's sloped plot lacks; or the mounds loosened and aerated the root zone, a mechanism independent of crop, soil, and slope that would help at home too. Because all four can account for the yield, the observation raises each likelihood only moderately, no explanation becomes compelling, and the household falls back on the default (Proposition 6). The farmer who concludes that no single hypothesis is supported is reasoning correctly, which offers one reading of why learning from own and others' experimentation is often slow and partial even where measured returns are high (Foster and Rosenzweig 1995; Conley and Udry 2010; Duflo et al. 2011; Emerick et al. 2016; BenYishay and Mobarak 2019). The framework also warns that the intuitive fix makes matters worse: supplying more reasons the technique might work inflates the field of rival explanations and stifles experimentation further (Proposition 7).

LtL promotes deep scientific understanding of the practices SBAE promotes, achieving cognitive dimension reduction on both margins. Students learn to explain why a technique works, its mechanism (Sankar et al. 2026), so the farmer does not treat each crop, soil, and slope as a separate hypothesis, and the hypothesis space shrinks from mechanism–plot pairs to mechanisms; and they learn under what conditions it yields larger returns, so the household can diagnose whether the mechanism is relevant ("aeration helps wherever soil sets hard"). Fewer competitors in the denominator of (5) and evidence from more plots bearing on the same hypothesis both raise the bound, which is the fourth design principle applied to a hypothesis space the agent did not choose.

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